Orbital Dragons to Hunt Cryptids: Neural Networks, Quantum-Inspired Data Science, and the Busy Beaver Problem

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Abstract

The busy beaver value $BB(n)$ is the longest finite runtime of any n-state Turing machine started on a blank tape. $BB(5)$ was settled in 2024 with a computer-checked proof. $BB(6)$ is blocked by a handful of machines whose tapes secretly run Collatz-style arithmetic; nobody can prove whether they stop. This paper studies two of these machines and surveys the rest. The approach is empirical first: the machine's behavior is logged as a sequence of events, the event data embedded in logarithmic, simplex, and hyperbolic coordinates, and the dynamics found to be piecewise affine in a five-integer state. The description is then proven correct in the Lean proof assistant, down to the machine's individual steps and with a minimal axiom footprint. The result is a machine-checked theorem: if a simple five-integer recurrence never enters an explicit finite set of bad configurations, the machine never halts. The converse pressure is also proven: entering a bad configuration forces the machine to halt. The open problem is thereby moved, without loss, from a Turing machine to a short arithmetic recurrence. The renewal map of the recurrence is then derived in closed form and shown to have a width-five fatal window around the diagonal, and the two mechanical proof routes with computational evidence: no congruence invariant up to modulus 384 excludes the window, and the branch dynamics has exponential factor complexity, so it is not conjugate to a rotation. Measuring the recurrence over its whole seed space shows the set of dying seeds has natural density zero at the Borel–Cantelli rate ($\approx 12/X$), so the studied machines' seeds are typical survivors. Surveying the entire holdout set with a two-ended harness (forward orbit against backward halt cone, regimes fitted in between) classifies all 1,094 open six-state machines: zero halts, zero alignments; a fixed-dimension class holding the certified four and, found by a tenfold-deeper sweep, three new members with derived conditional laws; an *array class* whose state spaces grow in dimension, for whose leading member an exact word-rewrite calculus is constructed (a two-phase odometer with a block-trisection boundary rule) validated against 2×10^8 raw steps; and 307 single-law machines among 346 with regime structure. The program is framed on a verification spectrum whose proven endpoint is $\mathsf{MIP}^* = \mathsf{RE}$: kernel checking at the bottom, certified predictive emulation (this paper) in the middle, and entangled-prover verification of halting-type claims as the goal. Every empirical claim is validated against 10^9 raw machine steps, and the recurrence is tested for 10^8 iterations. For the theory of learning, the pipeline is representation learning carried to a proof, and it is offered as a benchmark: unbounded, exactly labeled, law-governed sequence data on which a learned representation has a provable ground truth, with a two-prover game that verifies the forecasts of any learned emulator at constant cost per check.

1 The problem, from zero

A Turing machine here is a finite table: six control states, a tape of 0s and 1s, and for each (state, read symbol) pair an instruction (write a bit, move left or right, change state) or nothing. A

machine with no instruction for its current situation halts. $BB(n)$ is the longest runtime among the n-state tables that eventually halt from the all-zero tape. The function is noncomputable [4, 3], so each value is won machine by machine: every table either halts (run it) or must be *proven* to run forever.

The bbchallenge collaboration settled $BB(5) = 47{,}176{,}870$ in the Rocq proof assistant, "the first determination of a new Busy Beaver value in over 40 years" [1, 2]. At six states a residual set of machines defeats all known automated deciders; the community calls them cryptids. They are hard for a concrete reason: their tapes implement iterated arithmetic of the same character as the Collatz ($3x{+}1$) map, where the strongest known result, Tao's theorem that "$\mathrm{Col}_{\min}(N) \le f(N)$ for almost all N" [5], concerns typical orbits, while busy beaver questions need one specific orbit followed forever.

The subject here is a mirror-image pair of six-state cryptids, called dossier-2 and dossier-3. Machine one, written `1RB1RC_1LC1LC_1RE0LD_1LB1RF_---1RA_0RD1RD` in the standard notation, is the following table. Each entry gives (bit written, head move, next state); the machine starts in state A on an all-zero tape.

state	reads 0	reads 1
A	1, R, B	1, R, C
B	1, L, C	1, L, C
C	1, R, E	0, L, D
D	1, L, B	1, R, F
E	**(none: halt)**	1, R, A
F	0, R, D	1, R, D

The single empty entry is the only way to halt: the machine must be in state E reading a 0. Working backward through the table, state E is only entered from state C reading a 0, so any halt must pass through the configuration "C reads 0 with another 0 to its right." Call this the *probe event*. The reader should hold on to this configuration: every theorem below is, in the end, a statement about it, and the whole paper is one long argument that the probe event never happens.

2 Method

The method has three stages: watch, compress, certify. One principle recurs throughout. Every empirical discovery must eventually be either proven exactly or discarded; nothing in the pipeline is permitted to remain approximately true.

Watch. Run the machine and record only the moments that matter. For these machines there is a natural clock: a recurring configuration called a spawn entry. Between two spawn entries the machine performs one bounded "episode" of work. Sampling at spawn entries turns a raw trajectory of 10^9 steps into a clean sequence of a few million events.

Compress. At every spawn entry the tape has the same form: a solid block of 1s with exactly two 0-cells ("holes") inside it, the head at the block's right end. A concrete instant of the certified orbit (machine at position r, dots are 0s):

$$\cdots 0\,0\,\underbrace{1\,1\,1\,1\,1\,1}_{a}\,\underset{h_1}{0}\,\underbrace{1\,1\,1\,1}_{b-1}\,\underset{h_2}{0}\,\underbrace{1\,1\,1\,1\,1\,1\,1}_{c-1}\,\underset{r}{\downarrow}\,0\,0\cdots$$

So the whole configuration is five integers: the block's endpoints l, r, the hole positions h_1, h_2, and a tag; write $a = h_1 - l$, $b = h_2 - h_1$, $c = r - h_2$ for the gaps. Between spawn entries the machine performs one of seven "episodes," each later certified with its exact step count:

episode	steps	effect on the tape
spawn	7	extend block by 1, dig a fresh hole at $r-2$
crawl	10 per cycle	the walking gait; moves 2 cells, carries a hole
cross	13	absorb a hole, re-dig it 4 cells right
terminate	13	fill a hole, plant it 6 right (the probe events)
fuse	19	merge adjacent holes, plant 9 right
edge	10 or 13	extend the left end by 1 or 3 (renewal)
sweep	2 per cell	return home; holes cross for free

For instance, the crawl is a fixed ten-step loop: the machine marches left three cells twice (writing $1, 1, 0$ each time), then backs up four; the net effect is two cells of travel and a 0 that the walker carries along three cells behind itself. The danger state C reads a 0 *only* inside terminate and fuse episodes, and there the theorems show the cell to its right always holds a 1, so the probe event cannot fire. Plotted in raw coordinates, the event data looks like growth curves with no visible law (Figure 1). Plotted in the right coordinates the same data is piecewise linear: componentwise $\log(1+x)$ straightens each regime, the projective simplex $(a, b, c)/(a+b+c)$ removes scale, and hyperbolic and spherical renderings of those two charts expose the orbit's nested renewal structure (Figure 2). Straight segments are one regime; corners are regime changes.

These coordinate choices are quantum-inspired data science in a precise, bounded sense, and each has published lineage. The log chart is a dequantization: in Viro's words, "on logarithmic paper some real algebraic curves look like smoothed broken lines" [8], and Litvinov states the limit exactly: "tropical mathematics can be treated as a result of the so-called Maslov dequantization of the traditional mathematics over numerical fields as the Planck constant $\hbar$ tends to zero taking imaginary values" [9]. The piecewise-linear regimes here are that classical limit, observed in a Turing machine's arithmetic. The simplex chart with its Fisher–Rao metric is the classical face of quantum state geometry, made literal by the amplitude map $p \mapsto \sqrt{p}$ onto the positive octant of the sphere. The hyperbolic chart follows the machine-learning use of negative curvature for hierarchies, which "allows us to learn parsimonious representations of symbolic data by simultaneously capturing hierarchy and similarity" [10]; the nested renewal generations are exactly such a hierarchy. Two further disciplines carried over from quantum simulation practice: the vector encodings of Collatz-like steps are *reversible* maps (the circuit-model requirement that every operation be invertible), which is what makes backward orbit search, the classical analogue of uncomputation, available for the halt cones of the census below; and candidate states are evolved in ensembles. The boundary is stated plainly: no quantum computation is performed anywhere in this pipeline, and no quantum advantage is claimed; "quantum-inspired" names the origin of the geometries and the disciplines, not a resource. The underlying view of a Turing machine as a dynamical system goes back to Moore, for whose generalized shifts "many simple questions about their dynamics are undecidable" [6], and to Kůrka [7].

Certify. The picture suggests a law; the law must then be proven about the actual machine, or it is only a picture. It is first validated empirically and fail-closed: a predictor that knows the conjectured law must reproduce every event, with exact positions and exact tape, over 10^9 raw steps per machine, and any miss aborts. It reproduces all of them. The law is then proven in Lean 4, using only the kernel's minimal axioms (propext, Quot.sound; no classical choice). The certified account consists of: seven theorems, one per episode type, each giving the episode's exact step count, exact output shape, and safety; two inductions that handle the episodes' unbounded middle parts; a concrete certificate for the machine's startup (235 steps for one machine, 2,044 for the other); and a transport theorem showing that a five-integer function metaStep, iterated from the startup state, predicts the machine forever.

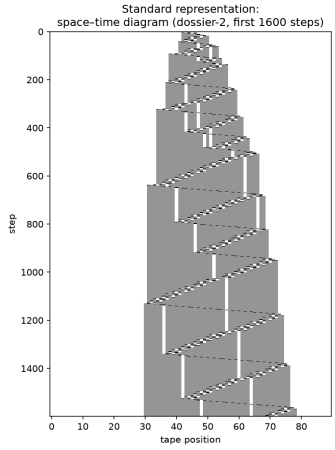


Figure 1: Raw view: the machine's first 1,600 steps (dark = 1s, one row per step). Nothing about the long-run law is visible here.

3 The result

Halting, for these machines, can only happen one way: a specific control state must read a 0 with another 0 beside it. The certified episode theorems show this situation never arises while the five-integer iteration stays inside an explicit guarded region. Writing SafeMetaOrbit for the statement that the iteration never leaves that region, and RawHalts(blankStart) for the statement that the machine halts from the all-zero tape:

Theorem 1. *For each machine, checked by the Lean kernel at axioms* {propext, Quot.sound}*:* SafeMetaOrbit $\to \neg$RawHalts(blankStart)*.*

Two facts make this reduction meaningful, not cosmetic. First, it is complete: startup, one exceptional (individually certified) episode, and every future step of the machine are covered by theorems, so nothing about the machine remains to be understood except the arithmetic. Second, it is sharp: the configurations just outside the guarded region were constructed, and the machine run from them; it halts within about fifty steps. Leaving the region is not merely unproven-safe. It is fatal. So SafeMetaOrbit *is* the halting problem, restated as arithmetic.

The arithmetic is explicit. Inside the main regime the iteration is $(a, b, c) \mapsto (a{+}6,\ b{-}2,\ c{-}2)$, so $\gamma = c - b$ is conserved while the counters drain. When b or c runs out, a renewal move re-seeds the triple by subtracting and swapping counters, with parity choosing the branch: part Euclidean algorithm, part Collatz. Figure 3 shows the renewal process as its own dynamical system: the return map of γ, with the fatal values $\gamma \in \{1, 3, 5\}$ marked. The recurrence avoids the band for 10^8 iterations (about 10^{15} machine steps). Proving it does so forever is the open problem, stated entirely in elementary arithmetic.

The problem admits a further characterization. Sampling the orbit at its renewal section (a Poincaré section of the dynamics) reduces the whole dynamics to a two-counter map, verified exactly against the certified orbit: with $(\alpha, \beta) = (6, a{-}1)$ or $(9, a{-}2)$ by the parity of a,

$$R(a,b) = \begin{cases} (a+6,\ b-3) & b \text{ even}, \\ (\alpha + 3(b-3),\ \beta - (b-3)) & b \text{ odd},\ b < \beta, \\ (\alpha + 3\beta + 3,\ b - \beta - 4) & b \text{ odd},\ b > \beta. \end{cases}$$

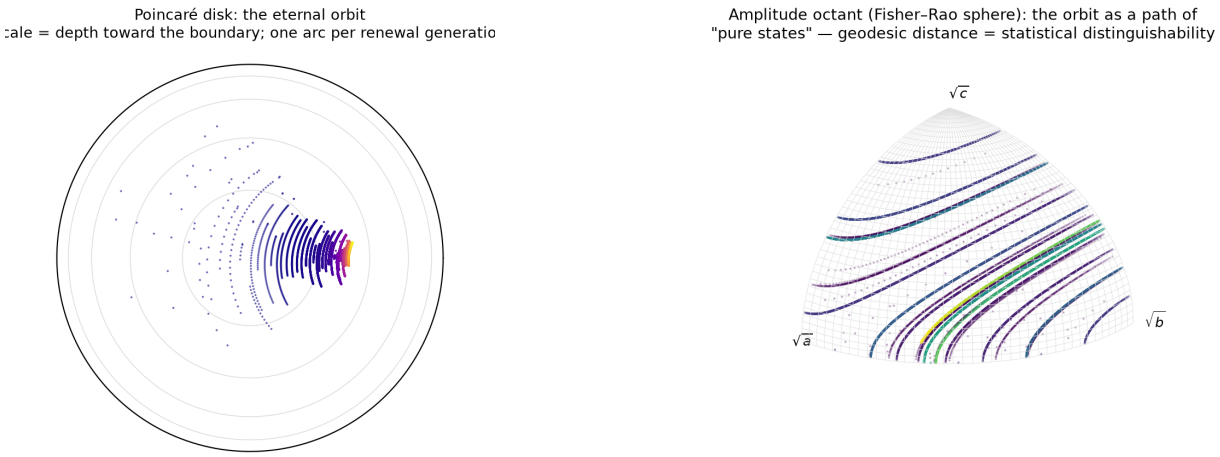


Figure 2: The same machine, 100,050 events, in two non-Euclidean coordinate systems. Left: the Poincaré disk. Shape runs along the boundary direction, log-scale is hyperbolic depth, and each renewal generation of the orbit sits one nested arc deeper. Right: the amplitude octant. The normalized state $(a, b, c)/(a+b+c)$ mapped by $p \mapsto \sqrt{p}$ onto the Fisher–Rao sphere, so the orbit becomes a path whose geodesic length measures statistical distinguishability. Straight arcs are single regimes; corners are regime changes.

The case $b = \beta$ never occurs on the verified orbit, against which the map is checked transition by transition. Its fatal set is geometrically simple: death occurs exactly when the two counters arrive almost equal ($b - \beta \in \{1, 3, 5\}$, plus two boundary cases). Two mechanical proof routes are then closed by computation. A sound residue-abstraction search finds a genuine mod-3 conservation law but shows that no congruence invariant up to modulus 384 excludes the fatal window. And the branch-symbol sequence has factor complexity growing like 1.6^n, where a rotation coding would give $n + 1$, so the dynamics is not conjugate to a rigid system, and continued-fraction rigidity arguments do not apply as stated. Over 10^5 renewals the orbit's minimum distance above the fatal window is 4: it grazes death and survives, so no proof can rely on a structural buffer. What remains is a pointwise avoidance statement for a pseudorandom multiplicative map, the Collatz situation, with three mitigations: the map is explicit and total, the fatal set is a finite union of lines near the diagonal, and the certified reduction stands under any future argument. The natural next language is hyperbolic: renewal steps are integer 2×2 matrix moves, and a near-tie of the counters is a cusp excursion on the modular surface, so the conjecture asks that a specific geodesic avoid three explicit horoballs forever [11].

Almost all seeds survive. The map R was also measured over its whole seed space, not just the certified orbit. Enumerating every seed in $[X, 2X)^2$ exactly for $X \le 256$, and sampling to $X = 2^{16}$, the fraction of seeds whose orbit ever dies is $\approx 12/X$: the dying set has natural density zero, at the rate a Borel–Cantelli heuristic predicts. The mechanism is visible in the data. Every observed death occurs within the first fifty renewals; the offset $b-\beta$ at the diagonal is uniformly distributed, fatal values included; and sizes grow by a factor $e^{0.134}$ per renewal, so the per-renewal hazard $\approx 6/\text{size}$ sums geometrically: death is an early-game phenomenon, and survival compounds. The studied machines' seeds are therefore *typical survivors*, not exceptions. (One measurement artifact is recorded here. An early instrument logged offsets only on steps actually taken and reported zero fatal offsets in 2×10^5 events; this is survivorship bias in textbook form, since a fatal offset is a death and never generates a step. Corrected instrumentation restored exact agreement with the

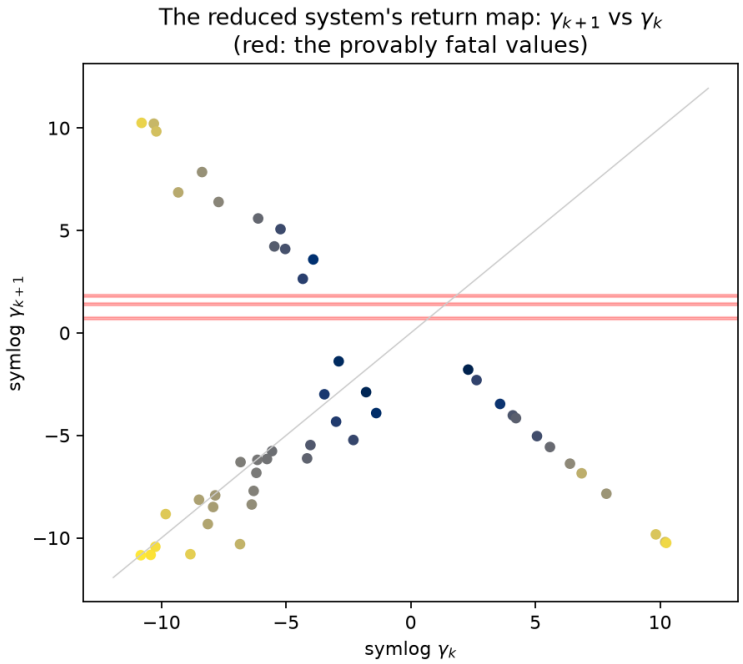


Figure 3: The reduced problem as a dynamical system: the return map of the renewal invariant, γ_{k+1} against γ_k (symlog axes), over the first fifty renewals. The red bands are the values proven fatal. The conjecture: the map's orbit never lands in a band.

uniform prediction.) A rigorous almost-all theorem needs one missing piece: the tail-hazard bound along orbits, an invariant-measure argument in the style of Tao's Collatz result, with the advantage that R is explicit, total, piecewise affine, and its hazard is measured uniform.

4 The census: what the remaining 1,094 machines are

The pipeline above certifies individual machines. Run in survey form across the entire BB(6) holdout set, it yields a classification: to the authors' knowledge, the first structural census of everything that remains open at six states.

The instrument is a three-part harness modeled on a two-prover frame (whose provenance the next section makes precise): a *start machine* (forward emulation from the blank tape), an *end machine* (the backward halt cone: every partial configuration that could precede a halt within k steps), and a *regime machine* that detects and fits the piecewise laws a run moves through, using second differences of extreme-event times. A machine halts if and only if its forward orbit ever aligns with its backward cone, so the verdict compiles from the three reports. Across all 1,094 holdouts: zero halts, and zero alignments; at every sampled instant, every holdout is at least 18 raw steps from any halting event. (For dossier-3 the two ends are provably incompatible in a stronger sense: all 134 backward families at depth 24 require three 0s within a six-cell window, a pattern no certified forward shape can produce; the statement is finite and checkable.) The census then splits the set:

- **4 machines** reduce to *fixed-dimension* integer maps, a bounded tuple of counters with certified episode theorems: the pair of this paper, an earlier machine of ours (dossier-1), and An-

tihydra, the community's best-known cryptid. These are the machines the previous sections certify. The initial ranked sweep found no others among its top 25; a subsequent tenfold-deeper sweep of the full structured pool falsified that boundary in the best direction, blindly re-deriving the certified four (with the smallest specs in the pool, an instrument validation) and surfacing *three new fixed-dimension machines*, each now carrying a derived conditional law validated fail-closed at 2×10^7 steps; the leading one is governed by five branch rules with a mod-3 divide branch, a new Collatz-like two-counter hybrid.

- **The array class.** Those 11 exact-but-unbounded machines are the new frontier: their tape shapes are growing *vectors* of blocks; the state space itself gains a dimension every few regimes. Their orbit specifications are exact but fragment into hundreds of rules, each witnessed by only one or two observed transitions; the law is real but lives on a space of growing dimension.
- **307 machines** are governed by a single affine law in the regime harness (the structured third of the cohort, 346 with detectable regime structure); the rest show no structure at the probed horizons.

For the top array-class machine (`1RB1RD_1LC0LC_ORD0LB_1RF0LE_ORD---_ORA1RC`) the array-class analogue of the episode calculus of the previous sections was then built: an exact *word-rewrite system*. Its tape at each canonical event is a word of 1-blocks of lengths 1, 2, 3 with single-0 gaps; a sweep of the head is a sequential transducer over that word, and the whole dynamics compresses to five interior rules (copy, absorb, cursor, insert, and one parametric regroup family with cost $8+6k$), two parametric boundary families, one of which is a literal *block trisection* ($3 \mapsto 1{,}1{,}1$), and two derived laws: a deposited 2-block acts as a *cursor*, restarting the next pass at its own position, and whether a pass ends in a restart is read off the output word. Semantically the machine is a two-phase odometer on $\{1, 3\}$-words: the 3s are carries, the cursor is the carry boundary. The system is validated the way the certified machines were before their Lean compilation. A *segment* is one full pass of the head between successive returns to the growing end of the tape; the words at those returns are the states of the rewrite system, and validation is out-of-sample: the rules are frozen after a training prefix and must reproduce every later segment. Rules learned on 4×10^7 raw steps reproduce 2,179 of 2,186 held-out segments over the next 1.6×10^8 steps *exactly* (output word, arrival type, step count, head position), with zero chaining violations; every exception is a flagged boundary-window event, never a silent misprediction, and re-learning over the full horizon leaves every decision key single-valued (zero conflicts across 3,956 segments). The sharpest check runs the machine itself on the calculus's own *predictions*: configurations forecast up to 139 sweeps beyond any simulated history, written onto a literal tape and handed to the raw machine, reproduce every forecast exactly, step counts included. This is predictive emulation (certify the law, then verify forecasts) standing, for the first time in this program, on a state space of growing dimension. The cohort boundary is sharp on both sides: a tenfold budget increase changes no verdict (all fourteen ranked candidates that failed the exact-spec gate still fail it, so the eleven-machine array class is complete under this criterion), and porting the formalism without machine-specific rules across the other ten machines places one at a 99.8% architecture match with 82.4% of held-out segments reproduced from literal rules alone, and two more at 99.3% and 98.6% exactness where extraction succeeds.

The classification also carries a quantitative signature. The abstract backward cone of dossier-3, the tree of renewal-level states dying within k steps, has level counts $9, 19, 31, 50, 81, \ldots$ satisfying $L(k) = L(k-1) + L(k-2)$ *exactly* from depth two on: a Fibonacci tree, branching at the golden ratio, and the same $\varphi \approx 1.618$ appears independently as the forward factor complexity. The mechanism is visible: a terminate step flips a parity, so no branch word contains two consecutive

terminates, precisely the constraint that generates Fibonacci counting. The standing, falsifiable prediction for the array class: each machine's death tree satisfies a small integer recurrence with a Pisot-like constant determined by its episode-alternation rules, the golden ratio being the founding case. The rank-1 machine supplies the second data point. Its cycle grammar inverts in closed form: the number of one-sweep preimages of an interior word equals $\prod_i 2^{\max(0,\, t_i - 2)}$ over its $(1, 3)$-train lengths t_i, so this death tree branches 2-adically in train lengths where dossier-3's branches at φ. Antihydra supplies the third: its abstract cone grows linearly (branching constant exactly 1, backward merges throttled by a counter guard), while $1/\varphi$ reappears in its seed-space hazard as the ruin root of $r^3 - 2r + 1$. Three machines, three constants, each derived from its episode-alternation rules; the golden ratio sits in one machine's cone and another machine's hazard.

The census closes on a consolidated ledger. Eighteen machines hold exact specifications of some grade. Six now carry kernel-checked conditional closures: the original four; the five-rule machine, whose theorem $\mathsf{SafeCheckpointOrbit} \to \neg\mathsf{RawHalts}$ is checked at the same two axioms with every episode of its timed law proven unconditionally except the divide renewal, folded into the conjecture and documented as such; and the rank-1 array-class machine itself, the first closure on a growing-dimension state space: $\mathsf{SafeWordOrbit} \to \neg\mathsf{RawHalts}$, in which every unit rule, both parametric boundary families, the mini-arrival mechanics, and the decision function are proven at cell-level grain with no assumption folded anywhere, the conjecture reducing to a single clause about a sequence of words. Three more machines hold fail-closed validated conditional laws awaiting the same compilation; each of the remaining nine carries either a law that is exact wherever extraction succeeds or a solved segmentation whose blocking step is named and bounded. Nothing in the exact-spec cohort remains uncharacterized; what remains is work. One negative result bounds that work from below: a sound invariant search (congruence closures to modulus 64, monotone integer functionals, verified in exact rational arithmetic, run by two independent implementations that first cross-audited each other's soundness) over every learned law in the cohort and over 258 single-law regime machines found no closed invariant excluding any fatal set. The holdout cohort has no invariant-decidable residue at that depth; conditional reduction is not a stopgap but the frontier.

5 The verification spectrum: where this sits, and the goal

This work sits on a spectrum of what "verifying a machine" can mean; both endpoints of the spectrum are now theorems.

Starting verification is the ground-up kind: replay and check every step. The Lean development here is this, industrialized: the kernel checks each of the 10^4 startup steps individually, and the trust surface is two axioms and a proof checker. It is complete and unconditional, but it can never outrun the machine: verifying n steps costs n.

Predictive emulation is the middle of the spectrum and the substance of this paper. Instead of replaying steps, certify the machine's *law*, the episode theorems and the five-integer map, and then verify *forecasts*: where the next probe event can occur, what the configuration will be 10^6 steps from now, whether halting is reachable from the current shape. One certified episode theorem covers unboundedly many raw steps; the certified recurrence covers all of them. Verification cost decouples from runtime. This is what lets two machines' entire futures be checked against one explicit arithmetic conjecture.

Ending verification is the horizon, taken here as the program's goal: $\mathsf{MIP}^* = \mathsf{RE}$ [12]. A classical polynomial-time verifier, interacting with two entangled provers, can verify *any* recursively enumerable statement, with the provers, not the verifier, bearing the unbounded cost; in the authors' words, "there is an efficient reduction from the Halting Problem to the problem of deciding whether

a two-player nonlocal game has entangled value 1 or at most 1/2" [12]. Non-halting statements such as ours live one level up (they are Π_1; zero-gap variants of MIP^* reach Π_2^0 [13]), so the entangled-prover frame in principle spans exactly the class of claims the busy beaver program generates.

This mapping is not only a framing; the certified reduction plays as an actual game. Define $G(K, r)$, verifying the claim "the machine halts within K renewals": two isolated provers commit to a history $(s_0, \ldots, s_K)$ of renewal states and a halt index k^*; the verifier checks the seed, samples r indices j and requires cross-consistency of the provers' snapshots together with $R(s_j) = s_{j+1}$ (one constant-cost arithmetic check each, which the certified theorems entitle in place of replay), and requires s_{k^*} to lie in the fatal set, whose members provably halt within about fifty raw steps. A halting machine gives honest provers a probability-1 win. Since R is deterministic with fixed seed, the only law-consistent history is the true orbit, which stays at distance at least 4 above the fatal window over the tested horizon, so any winning strategy needs a seam, caught per sample with probability $1/K$. Played at $K = 10^5$: the true orbit contains zero fatal states, so no false acceptance is possible on the real history; spliced forgeries are accepted at the rate the seam bound predicts (0.998 at $r = 64$, 0.745 at $r = 2^{14}$, 0 in all trials at $r = 2^{18}$ and 2^{20}, where the analytic bound is below 3×10^{-5}). The verifier's total work is r integer checks on five-integer states whose raw-tape realizations at that depth are astronomically long: verification cost decoupled from runtime, played as a game. The same game runs unchanged over the array-class word calculus, with one rewrite-system evaluation as the law check and raw execution of the materialized word as the terminal check; forged terminals built from the machine's own depth-24 halt cone (stamped into true orbit words of about 10^4 cells) genuinely halt, and are then caught by law sampling at the identical seam rate, 0 acceptances in all trials at $r \geq 2048$ over a 346-segment word history. At the certified five-integer layer the terminal check is raw execution through the holeTape realization, and the lethality theorem is observed live: every constructed danger-band corner state halts within 6 to 12 raw steps, while 20,000 random hole-perturbations of true states produce no halting configuration at all. Three state-space classes, one verifier.

The alignment is this: kernel verification anchors truth at the bottom; predictive emulation compresses it in the middle; and $\mathsf{MIP}^* = \mathsf{RE}$ says that at the top, verification of halting-type claims is possible *in principle* with entanglement as the resource. The quantum-inspired toolkit above (reversible vector maps, state-space geometry) borrows disciplines from that endpoint; the open engineering-of-mathematics question is how much of the middle rung, certified forecasts, can be pushed toward it, replacing replay with interaction. Nothing in the results depends on this framing; it states the destination the pipeline is pointed at.

6 A benchmark for the theory of learning

There are three reasons this may interest this community. The first is the workflow. This project is an instance of the emerging human–AI–kernel pattern in mathematics, in which AI systems draft and compile proofs at scale, a proof kernel arbitrates, and humans own claims; this is the pattern of Tao's Equational Theories Project, which "successfully determined all 22 028 942 edges of the implication graph between the 4694 simplest equational laws on magmas," and of the new Palomar registry for Lean-verified results [14, 15]. All of the Lean code was drafted by an AI assistant from validated empirical specifications; the kernel checked roughly fifteen thousand lines; the generators are convenience, not trust surface. The second reason is methodological kinship. Finding that a high-dimensional trajectory is governed by a low-dimensional piecewise-affine law, by choosing coordinates in which the structure is linear, is the daily business of representation learning. Here the same move is carried one step further: the discovered representation is not evaluated on held-out

data but proven exactly correct, step by step, against the system that generated it. Each pipeline stage has a standard learning-theory name. Event extraction is feature engineering; the coordinate charts are the representation; the piecewise-affine law is a linear model in the right representation; fail-closed validation is a generalization test with a zero-tolerance loss; the kernel proof replaces the test set.

The third reason is a concrete offer. Busy beaver machines generate unbounded, exactly labeled, law-governed sequence data, and the census grades them: fixed-dimension machines whose ground-truth representation is a proven five-integer state, array-class machines whose ground truth is a word-rewrite system of growing dimension, and a structured remainder with fitted regime laws. A network trained to emulate one of these machines can be scored not against samples but against certainty: slot it in as prover A of the halting game and every forecast it makes is checked at constant cost, with any deviation from the certified law caught at the seam rate. What a feature learner should discover here is known exactly, chart by chart, which is precisely the situation the mathematical theory of deep networks lacks in natural data.

Scope. No unconditional non-halting claim is made. The coordinate charts guide discovery and exposition; no theorem depends on them. Empirical horizons are what was run, not what is true.

Acknowledgments. The Lean developments, empirical validation, and proof generators were built in collaboration with an AI assistant (Claude, Anthropic). Every proof is checked by the Lean 4 kernel at the stated axioms.

References

[1] The bbchallenge Collaboration. Determination of the fifth Busy Beaver value. arXiv:2509.12337. "Our result marks the first determination of a new Busy Beaver value in over 40 years and the first Busy Beaver value ever to be formally verified." The kernel-checked standard the Lean developments here follow; source of the BB(6) holdout set the census classifies.

[2] The bbchallenge Collaboration. Turing machine deciders, part I. arXiv:2504.20563. "Each decider is specialised in recognising a particular type of non-halting behavior." The automated paradigm whose residue at six states is the 1,094-machine cohort studied here.

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